

codecs that are the best for at least one dataset (7 different)

common defaults

Simulation

Experiment

Analysis outputs

Graphs

Deep learning

	SPratio	ANS +4-byte shuffle	Bitcomp sparse int32	ANS float16	ANS +2-byte shuffle	SPratio +bit shuffle	SPspeed +2-byte shuffle	Zstd	LZ4
Nyx (cosmology)	best	74	139	283	105	107	108	357	206
VPIC (plasma)	18	best	21	9	7	2	14	179	85
WarpX (plasma)	best	8	18	20	14	8	12	178	72
LAMMPS (molecular)	9	8	12	9	11	10	best	158	59
CAMELS (galaxies)	best	0	17	10	8	3	12	187	99
Nyx analysis	best	397	144	2.2k	495	89	397	1.3k	1.4k
VPIC analysis	1	13	best	306	25	38	160	906	908
LAMMPS analysis	best	43	36	162	68	16	91	1.4k	1.1k
CMS NanoAOD	69	best	48	96	17	7	92	633	521
Genome reads	90	best	58	61	0	38	102	429	411
X-ray detector	11	10	best	290	15	63	185	1.1k	1.4k
FEM matrices	best	7	13	16	12	11	18	103	43
LiveJournal	best	18	31	37	30	11	29	305	193
Orkut	best	48	49	82	68	29	66	428	295
ResNet-18 training	45	best	43	30	5	54	62	262	143
Pythia checkpoints	48	1	50	19	10	best	42	252	142
Qwen bf16 weights	37	4	40	best	4	6	30	202	98
OPT activations	115	1	38	234	best	296	425	539	536
GPT-2 KV cache	11	4	13	best	4	10	10	173	62
mean extra cost (%)	24	33	41	206	47	42	98	479	408

cell = extra cost (%) of using the column's codec for the whole row's dataset, vs that dataset's own best codec

 the dataset's own best
 < 10
 10-25
 25-50
 50-100
 100-300
 > 300